

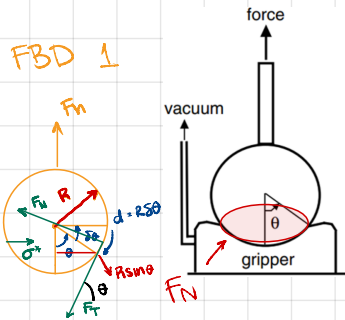
Problem 1 Part A Corrections made w/ this color

Assumptions

- * uniform contraction when vacuum is applied
- * pinching occurs in a thin band of width d near the contact line
- * pinched region acts like an o-ring pressing against the sphere
- * the stress σ^* is uniform in the pinching region
- * the sphere is rigid compare to the gripper material
- * friction follows Coulomb's law w/ coefficient μ

Variables

- * R = sphere radius
- * θ = contact angle (angle from vertical to contact line)
- * μ = coefficient of friction
- * σ^* = pinching stress in the O-ring region
- * d = width of pinching region
- * A_0 = area of the pinching region
- * A^* = horizontal cross-sec area inside contact line



① Porous Sphere (Friction only)

A pinching region forms an O-ring around the sphere at angle θ
 $\Rightarrow A_0 = 2\pi R d \sin \theta$

Normal force is the stress times area times the component normal to the surface

$$\Rightarrow F_N = \sigma^* A_0 \sin \theta = \sigma^* (2\pi R d \sin \theta) \sin \theta = 2\pi R d \sigma^* \sin^2 \theta$$

$F_f = \mu F_N$, only the vertical component contributes to holding

$$\therefore F_f = F_N (\mu \sin \theta - \cos \theta) = 2\pi R d \sigma^* \sin^2 \theta (\mu \sin \theta - \cos \theta)$$

Note: gripping only occurs when $\mu \sin \theta - \cos \theta > 0 \Rightarrow \theta > \theta_c = \arctan\left(\frac{1}{\mu}\right)$

$$\therefore F_h = 2\pi R d \sigma^* (\mu \sin \theta - \cos \theta) \sin^2 \theta$$

② Solid sphere (Friction + Suction)

For a sphere, the horizontal cross-sectional area inside the contact line is:

$$A^* = \pi R^2 \sin^2 \theta$$

The suction force is $F_s = P_g A^*$, where P_g is the gap pressure. Relating P_g to pinching stress

$$P_g = \frac{F_f}{A_0} = \sigma^* \sin \theta (\mu \sin \theta - \cos \theta)$$

$$\Rightarrow F_s = P_g A^* = \pi R^2 \sigma^* (\mu \sin \theta - \cos \theta) \sin^3 \theta$$

$$\therefore F_h = F_s + F_f = \pi R^2 \sigma^* (\mu \sin \theta - \cos \theta) \sin^3 \theta + 2\pi R d \sigma^* (\mu \sin \theta - \cos \theta) \sin^2 \theta$$

$$F_h = \pi R^2 \sigma^* (\mu \sin \theta - \cos \theta) \sin^3 \theta \left(1 + \frac{2d}{R \sin \theta} \right)$$

Unclear details from research paper:

- It does not explain how pinching width d is determined
- How σ^* relates to P_{jam} is not quantitatively derived
- The transition between bending-dominated and stretching-dominated interlocking for $\theta > \pi/2$ lacks criteria

Part B Look at code!! Plot holding force vs contact angle

Part C // Plot critical angle vs friction coefficient

As μ increases, θ_c decreases significantly. A lower θ_c means the gripper can hold objects with shallower contact angles. Gripping begins at smaller deformation angles. It can also grip flatter objects and less precise positioning required

Part D θ_c decreases 6.8° , holding force increase by 25-30% across all contact angles. Therefore, gripper can engage earlier with shallower contact, and more robust gripping of objects with low curvature

Part E Alternative Gripper Design

Adding micro-scale patterns or textures to the gripper membrane surface to enhance geometric interlocking at smaller angles

Expected Effects on F_h (holding force)

- texture increases effective μ through mechanical interlocking
- better conformal contact for vacuum sealing
- small-scale geometric constraints

Modified Model $M_{eff} = M_{adhesive} + M_{texture}$

Mtexture depends on:

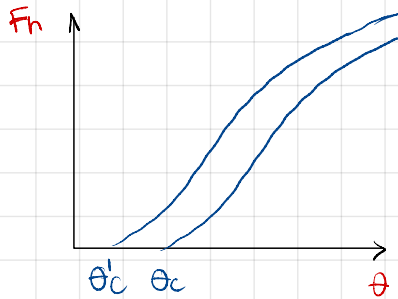
→ texture geometry (spacing, shape, height)

→ material compliance

→ Local curvature of object surface

Modified Equation $\Rightarrow F_n = \pi R^2 \sigma^* (M_{eff} \sin \theta - \cos \theta) \sin^3 \theta \left(1 + \frac{2d_{eff}}{R \sin \theta} \right)$ ✓

where d_{eff} may increase due to better conformal contact



We can get:

* Lower critical angle

* Higher maximum holding force

Check code for plots !!

Reasoning: Micro-scale features can engage with object surface roughness. Texture provides additional gripping mechanisms before full macroscopic contact is established. Lastly, the modification would be most beneficial for smooth, hard to grip surfaces

 Open in Colab

Problem 1 - Part B

```
In [11]: import numpy as np
import matplotlib.pyplot as plt

# Given parameters
mu = 1.1
sigma_star = 54e3 # Pa
d = 1.07e-3 # m
R = 19e-3 # m

# Critical angle
theta_c = np.arctan(1/mu)

# Create theta range from theta_c to pi/2
theta = np.linspace(theta_c, np.pi/2, 100)

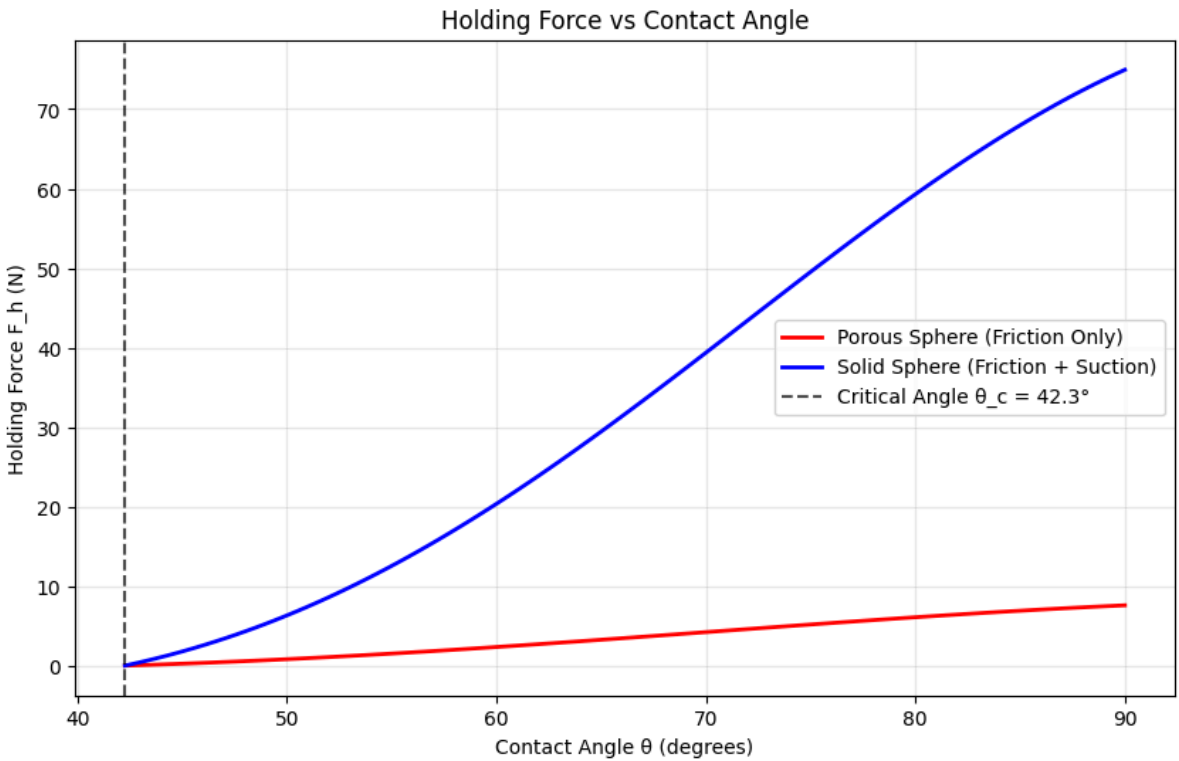
# Porous sphere (friction only)
F_h_porous = 2 * np.pi * R * d * sigma_star * (mu * np.sin(theta) - np.cos(theta))

# Solid sphere (friction + suction)
F_h_solid = (np.pi * R**2 * sigma_star * (mu * np.sin(theta) - np.cos(theta)) *
            (1 + (2*d)/(R * np.sin(theta))))

# Convert to Newtons and plot
plt.figure(figsize=(10, 6))
plt.plot(np.degrees(theta), F_h_porous, 'r-', linewidth=2, label='Porous Sphere')
plt.plot(np.degrees(theta), F_h_solid, 'b-', linewidth=2, label='Solid Sphere (Friction + Suction)')
plt.axvline(np.degrees(theta_c), color='k', linestyle='--', alpha=0.7, label=f'Critical Angle θ_c = {np.degrees(theta_c):.2f}°')

plt.xlabel('Contact Angle θ (degrees)')
plt.ylabel('Holding Force F_h (N)')
plt.title('Holding Force vs Contact Angle')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()

print(f"Critical angle θ_c = {np.degrees(theta_c):.2f}°")
print(f"Maximum holding force (porous): {np.max(F_h_porous):.2f} N")
print(f"Maximum holding force (solid): {np.max(F_h_solid):.2f} N")
```



Critical angle $\theta_c = 42.27^\circ$
Maximum holding force (porous): 7.59 N
Maximum holding force (solid): 74.95 N

Problem 1 - Part C

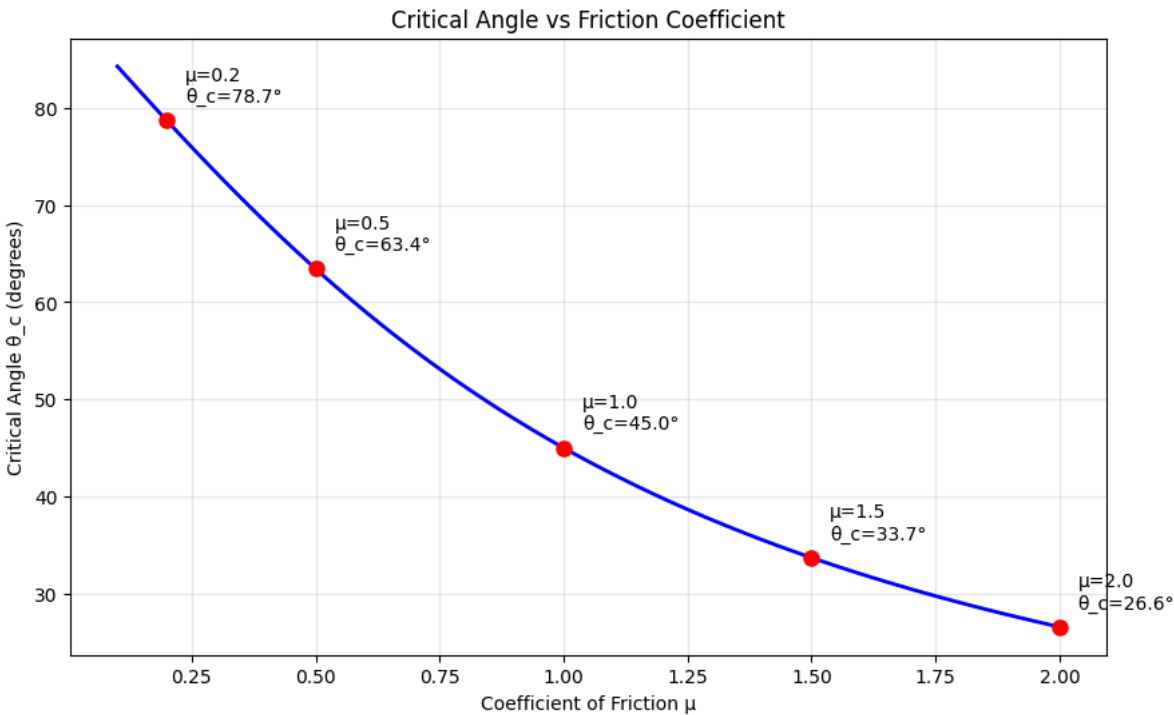
```
In [12]: import numpy as np
import matplotlib.pyplot as plt

# Physically achievable range of friction coefficients
# Rubber on various surfaces can achieve  $\mu = 0.1$  to  $2.0+$ 
mu_range = np.linspace(0.1, 2.0, 100)
theta_c_range = np.arctan(1/mu_range)

plt.figure(figsize=(10, 6))
plt.plot(mu_range, np.degrees(theta_c_range), 'b-', linewidth=2)
plt.xlabel('Coefficient of Friction  $\mu$ ')
plt.ylabel('Critical Angle  $\theta_c$  (degrees)')
plt.title('Critical Angle vs Friction Coefficient')
plt.grid(True, alpha=0.3)

# Mark some realistic values
realistic_mu = [0.2, 0.5, 1.0, 1.5, 2.0]
for mu_val in realistic_mu:
    theta_val = np.degrees(np.arctan(1/mu_val))
    plt.plot(mu_val, theta_val, 'ro', markersize=8)
    plt.annotate(f' $\mu={mu\_val}$ \n $\theta_c={theta\_val:.1f}^\circ$ ',
                (mu_val, theta_val),
                xytext=(10, 10), textcoords='offset points')

plt.show()
```



Problem 1 - Part D

```
In [3]: # Parameters with increased friction
mu_high = 1.4
sigma_star = 54e3 # Pa
d = 1.07e-3 # m
R = 19e-3 # m

# New critical angle
theta_c_high = np.arctan(1/mu_high)

# Theta range from new critical angle to  $\pi/2$ 
theta_high = np.linspace(theta_c_high, np.pi/2, 100)

# Calculate holding forces
F_h_porous_high = 2 * np.pi * R * d * sigma_star * (mu_high * np.sin(theta_high))
F_h_solid_high = (np.pi * R**2 * sigma_star * (mu_high * np.sin(theta_high) - np
                (1 + (2*d)/(R * np.sin(theta_high)))))

# Compare with original  $\mu = 1.1$ 
```



```
mu_orig = 1.1
theta_c_orig = np.arctan(1/mu_orig)
theta_orig = np.linspace(theta_c_orig, np.pi/2, 100)

F_h_porous_orig = 2 * np.pi * R * d * sigma_star * (mu_orig * np.sin(theta_orig)
F_h_solid_orig = (np.pi * R**2 * sigma_star * (mu_orig * np.sin(theta_orig) - n
                (1 + (2*d)/(R * np.sin(theta_orig))))

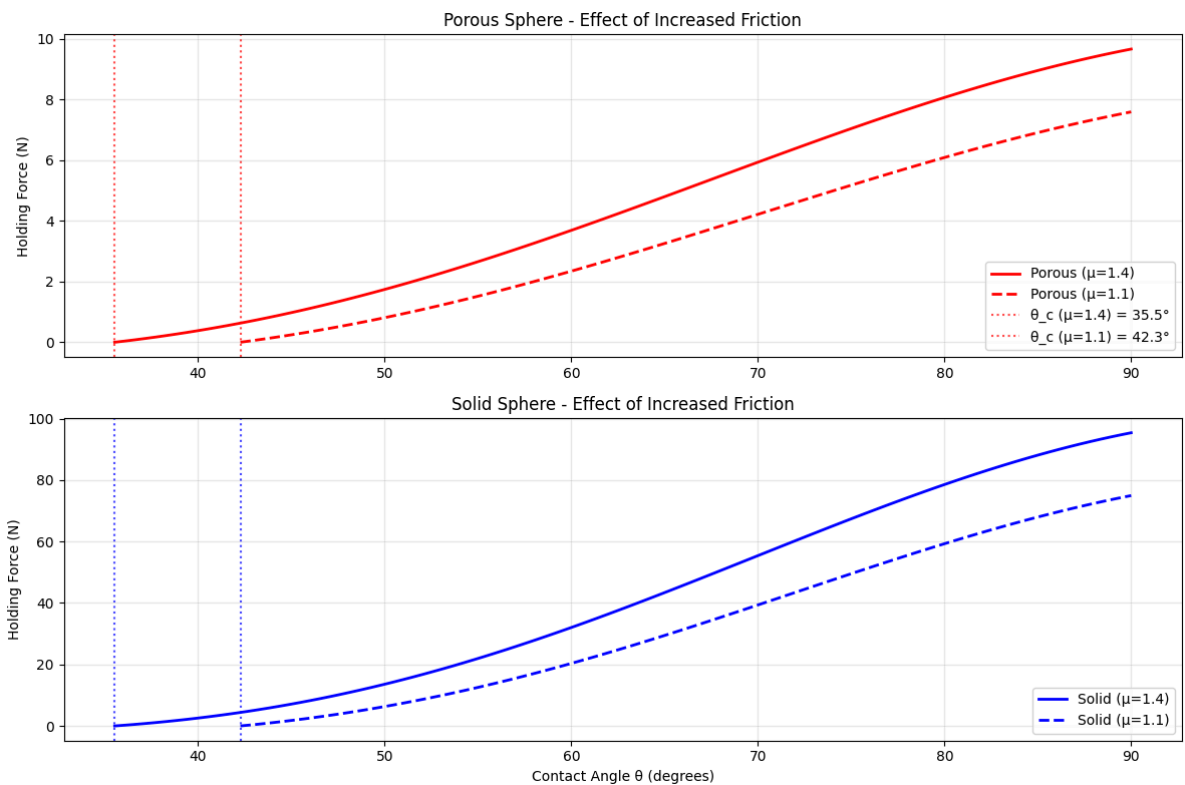
# Plot comparison
plt.figure(figsize=(12, 8))

plt.subplot(2, 1, 1)
plt.plot(np.degrees(theta_high), F_h_porous_high, 'r-', linewidth=2, label='Porous')
plt.plot(np.degrees(theta_orig), F_h_porous_orig, 'r--', linewidth=2, label='Porous')
plt.axvline(np.degrees(theta_c_high), color='r', linestyle=':', alpha=0.7, label='θ_c (μ=1.4)')
plt.axvline(np.degrees(theta_c_orig), color='r', linestyle=':', alpha=0.7, label='θ_c (μ=1.1)')
plt.ylabel('Holding Force (N)')
plt.title('Porous Sphere - Effect of Increased Friction')
plt.legend()
plt.grid(True, alpha=0.3)

plt.subplot(2, 1, 2)
plt.plot(np.degrees(theta_high), F_h_solid_high, 'b-', linewidth=2, label='Solid')
plt.plot(np.degrees(theta_orig), F_h_solid_orig, 'b--', linewidth=2, label='Solid')
plt.axvline(np.degrees(theta_c_high), color='b', linestyle=':', alpha=0.7)
plt.axvline(np.degrees(theta_c_orig), color='b', linestyle=':', alpha=0.7)
plt.xlabel('Contact Angle θ (degrees)')
plt.ylabel('Holding Force (N)')
plt.title('Solid Sphere - Effect of Increased Friction')
plt.legend()
plt.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

print(f"Critical angle reduction: {np.degrees(theta_c_orig - theta_c_high):.1f}°")
print(f"Maximum force increase (porous): {(np.max(F_h_porous_high)/np.max(F_h_porous_orig)):.1%}")
print(f"Maximum force increase (solid): {(np.max(F_h_solid_high)/np.max(F_h_solid_orig)):.1%}")
```



Critical angle reduction: 6.7°
Maximum force increase (porous): 27.3%
Maximum force increase (solid): 27.3%

```
In [2]: import numpy as np
import matplotlib.pyplot as plt

# Base parameters from the paper
sigma_star = 54e3 # Pa
```

```

d = 1.07e-3 # m
R = 19e-3 # m
mu_original = 1.4

# Textured membrane parameters
mu_textured = 1.4 # Increased effective friction
d_effective = 1.3e-3 # Increased effective contact width due to better sealing

def holding_force_porous(theta, mu, d_eff):
    """Holding force for porous sphere (friction only)"""
    return 2 * np.pi * R * d_eff * sigma_star * (mu * np.sin(theta) - np.cos(theta))

def holding_force_solid(theta, mu, d_eff):
    """Holding force for solid sphere (friction + suction)"""
    return (np.pi * R**2 * sigma_star * (mu * np.sin(theta) - np.cos(theta)) * (1 + (2*d_eff)/(R * np.sin(theta))))

# Calculate critical angles
theta_c_original = np.arctan(1/mu_original)
theta_c_textured = np.arctan(1/mu_textured)

# Create theta ranges
theta_original = np.linspace(theta_c_original, np.pi/2, 100)
theta_textured = np.linspace(theta_c_textured, np.pi/2, 100)

# Calculate holding forces
F_porous_orig = holding_force_porous(theta_original, mu_original, d)
F_solid_orig = holding_force_solid(theta_original, mu_original, d)

F_porous_text = holding_force_porous(theta_textured, mu_textured, d_effective)
F_solid_text = holding_force_solid(theta_textured, mu_textured, d_effective)

# Plot comparison
plt.figure(figsize=(14, 10))

# Porous sphere comparison
plt.subplot(2, 2, 1)
plt.plot(np.degrees(theta_original), F_porous_orig, 'r-', linewidth=3, label='Original')
plt.plot(np.degrees(theta_textured), F_porous_text, 'r--', linewidth=3, label='Textured')
plt.axvline(np.degrees(theta_c_original), color='r', linestyle=':', alpha=0.7,
             label=f'Original  $\theta_c = \{{np.degrees(theta_c\_original):.1f}\}^\circ$ ')
plt.axvline(np.degrees(theta_c_textured), color='r', linestyle=':', alpha=0.7,
             label=f'Textured  $\theta_c = \{{np.degrees(theta_c\_textured):.1f}\}^\circ$ ')
plt.ylabel('Holding Force (N)')
plt.title('Porous Sphere - Textured vs Original Membrane')
plt.legend()
plt.grid(True, alpha=0.3)

# Solid sphere comparison
plt.subplot(2, 2, 2)
plt.plot(np.degrees(theta_original), F_solid_orig, 'b-', linewidth=3, label='Original')
plt.plot(np.degrees(theta_textured), F_solid_text, 'b--', linewidth=3, label='Textured')
plt.axvline(np.degrees(theta_c_original), color='b', linestyle=':', alpha=0.7)
plt.axvline(np.degrees(theta_c_textured), color='b', linestyle=':', alpha=0.7)
plt.ylabel('Holding Force (N)')
plt.title('Solid Sphere - Textured vs Original Membrane')
plt.legend()
plt.grid(True, alpha=0.3)

# Force enhancement percentage
theta_common = np.linspace(max(theta_c_original, theta_c_textured), np.pi/2, 100)
F_porous_orig_common = holding_force_porous(theta_common, mu_original, d)
F_porous_text_common = holding_force_porous(theta_common, mu_textured, d_effective)
F_solid_orig_common = holding_force_solid(theta_common, mu_original, d)
F_solid_text_common = holding_force_solid(theta_common, mu_textured, d_effective)

plt.tight_layout()
plt.show()

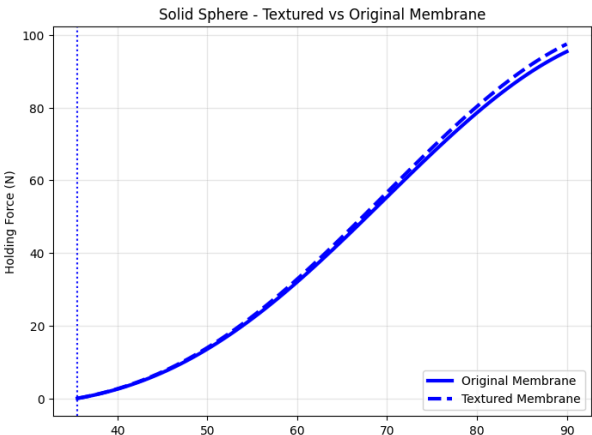
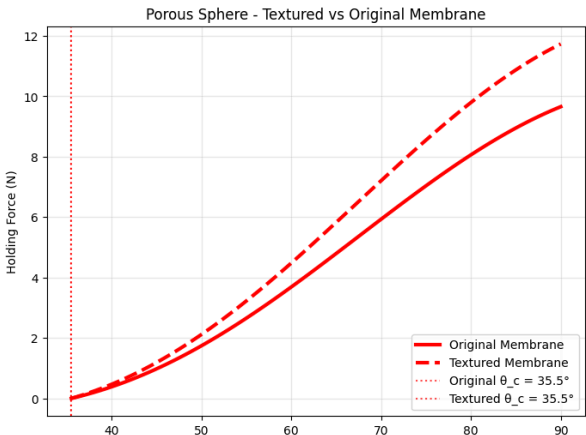
# Quantitative analysis
print("=== TEXTURED MEMBRANE EFFECTS ===")
print(f"Critical angle reduction: {np.degrees(theta_c_original - theta_c_textured):.1f}°")
print(f"Effective contact width increase: {(d_effective/d - 1)*100:.0f}%")

```

```
max_theta = np.pi/2
F_porous_max_orig = holding_force_porous(max_theta, mu_original, d)
F_porous_max_text = holding_force_porous(max_theta, mu_textured, d_effective)
F_solid_max_orig = holding_force_solid(max_theta, mu_original, d)
F_solid_max_text = holding_force_solid(max_theta, mu_textured, d_effective)

print(f"\nMaximum holding force at  $\theta = 90^\circ$ :")
print(f"Porous sphere - Original: {F_porous_max_orig:.2f} N")
print(f"Porous sphere - Textured: {F_porous_max_text:.2f} N")

print(f"\nSolid sphere - Original: {F_solid_max_orig:.2f} N")
print(f"Solid sphere - Textured: {F_solid_max_text:.2f} N")
```



=== TEXTURED MEMBRANE EFFECTS ===
Critical angle reduction: 0.0°
Effective contact width increase: 21%

Maximum holding force at $\theta = 90^\circ$:
Porous sphere - Original: 9.66 N
Porous sphere - Textured: 11.73 N

Solid sphere - Original: 95.40 N
Solid sphere - Textured: 97.47 N